BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Secondary Storage and Device Choice

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. Previous lesson: which memory is volatile, RAM or ROM?

2. Why does a CPU use cache?

3. Convert 6400 bits into bytes.

4. Why does a saved SSD file survive power-off but unsaved RAM data may not?

Today's objectives

Compare magnetic, optical and solid-state storage; justify a suitable choice.

Key vocabulary - Read, cover, recall

magnetic
Bits recorded using magnetism

optical
Disc read using laser light

solid-state
Memory chips without moving heads

secondary
Stores files without power

Write one definition in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Non-volatile secondary storage

Saved files remain when the computer is switched off.

Compare technologies using the needs of the actual user.

MAGNETIC HDD	OPTICAL DISC	SOLID-STATE
Magnetic patterns	Laser reads disc	Memory chips
Moving parts	Removable media	No moving heads
Low cost per GB	**Useful for distribution**	**Fast, portable**

No one technology is best for every use.

I Do - Recommend with evidence

Scenario: A travelling photographer moves large files.

Need 1: fast transfers. Need 2: resistance to knocks.

Recommendation: portable solid-state drive (SSD).

Reason: no moving heads; generally quick file transfers.

Trade-off: SSDs may cost more per GB than HDDs.

We Do - Match requirements

Choose a suitable type of secondary storage, then justify it.

Large school archive, low cost: _____

Portable storage for homework: _____

Disc read using a laser: _____

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

Start on paper. Check the model answers only on page 4.

```
capacityGB = 64    fileGB = 8
print(capacityGB / fileGB)
```

Predict how many 8 GB files fit into 64 GB.

Change fileGB to 16 and find the new answer.

Why does capacity alone not decide the best device?

You Do - Independent work

Name two secondary storage technologies. [2]

A travelling photographer drops their bag: choose and justify a device. [3]

Why may an HDD suit a large low-cost school archive? [3]

Thinking: explain why a fast SSD is not the best option for every task.

Exit Ticket - True or False?

Answer before you turn to the green-pen page.

1. Secondary storage retains data after power is removed. ☐ T ☐ F
2. An SSD needs a moving read/write head. ☐ T ☐ F
3. Storage is chosen using more than speed alone. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick correct answers. For a mistake, use the model answer then copy the FULL question and write a fresh answer below.

Do Now - Answers

Q1 RAM. ☐

Q2 Fast access to often-needed data. ☐

Q3 800 bytes. ☐

Q4 SSD is non-volatile; RAM is volatile. ☐

We Do - Guided practice

Archive: HDD; portable: SSD/USB; optical: disc

Key vocabulary

magnetic ☐
Bits recorded using magnetism

optical ☐
Disc read using laser light

solid-state ☐
Memory chips without moving heads

secondary ☐
Stores files without power

Code First - Answers

Predict ☐
64 / 8 = 8 files

Write ☐
64 / 16 = 4 files

Improve ☐
Also consider speed, cost, portability and durability

Exit Ticket - Explanations

1. True

2. False: SSD uses memory chips

3. True

You Do - Self-mark

Q1	Any two: magnetic, optical, solid-state.	☐
Q2	Portable SSD: no moving heads; faster transfers; less knock damage.	☐
Q3	HDD: high capacity and relatively low cost per GB.	☐
Thinking	It may cost more per GB; some tasks prioritise capacity or price.	☐

Try the question again

1. Copy the FULL question you got wrong (or the hardest):

2. In GREEN PEN, write a NEW answer. Show working: